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1. Preparation of alkanes from catalytic reduction of unsaturated hydrocarbons.

When a mixture hydrogen gas with alkene or alkyne gas is passed over a catalysts such as platinum or palladium at room temperature, an alkane is produced. This process of addition of H_2 to unsaturated compounds is known as hydrogenationThe above process can be catalysed by nickel at 298K. This reaction is known as Sabatier-Sendersens reaction

for example:

$$\underset{\text{propene}}{CH_3-CH=CH_2} + H_2 \xrightarrow{Pt} \underset{\text{propane}}{CH_3-CH_2-CH_3}$$

$$\underset{\text{ethene}}{CH_2=CH_2} + H_2 \xrightarrow[298K]{Ni} \underset{\text{ethane}}{CH_3-CH_3}$$

$$\underset{\text{prop-1-yne}}{CH_3-C\equiv CH} + 2H_2 \xrightarrow{Pt} \underset{\text{propane}}{CH_3-CH_2-CH_3}$$

2. Preparation of alkanes from carboxylic acids:

i) Decarboxylation of sodium salt of carboxylic acid

When a mixture of sodium salt of carboxylic acid and soda lime (sodium hydroxide + calcium oxide) is heated, alkane is formed. The alkane formed has one carbon atom less than carboxylic acid. This process of eliminating carboxylic group is known as decarboxylation.

for example:

$$\underset{\text{Sodium acetate}}{CH_3COONa} + NaOH \xrightarrow[\Delta]{CaO} \underset{\text{Methane}}{CH_4} + Na_2CO_3$$

ii) Kolbe's Electrolytic method

When sodium or potassium salt of carboxylic acid is electrolyzed, a higher alkane is formed. The decarboxylative dimerization of two carboxylic acid occurs. This method is suitable for preparing symmetrical alkanes(R-R).

$$2CH_3COONa + 2H_2O \xrightarrow{\text{Electrolysis}} \underbrace{H_3C-CH_3}_{\text{at Anode}} + 2CO_2 + \underbrace{H_2 + 2NaOH}_{\text{at Cathode}}$$

3.Preparation of alkanes using alkyl halides (or) halo alkanes

i) By reduction with nascent hydrogen

Except alkyl fluorides, other alkyl halides can be converted to alkanes by reduction with nascent hydrogen. The hydrogen for reduction may be obtained by using any of the following reducing agents: Zn+HCl, Zn+CH_3COOH, Zn-Cu couple in ethanol, $LiAlH_4$ etc.,

for example:

$$\underset{\text{chloro propane}}{CH_3-CH_2-CH_2-Cl} \xrightarrow[Zn/HCl]{[H]} \underset{\text{propane}}{CH_3-CH_2-CH_3} + HCl$$

ii) Wurtz reaction

When a solution of halo alkanes in dry ether is treated with sodium metal,